

30) Let V be the center of mass of a system of point masses located at $V_1, \dots, V_K$ as in Exercise 29. Is V in $\text{Span}\{V_1, \dots, V_K\}$? Explain.

Solution: From Ex-29, the center of mass of the system is

$$\bar{V} = \frac{1}{m} [m_1 V_1 + \dots + m_K V_K] = \left(\frac{m_1}{m}\right) V_1 + \dots + \left(\frac{m_K}{m}\right) V_K.$$

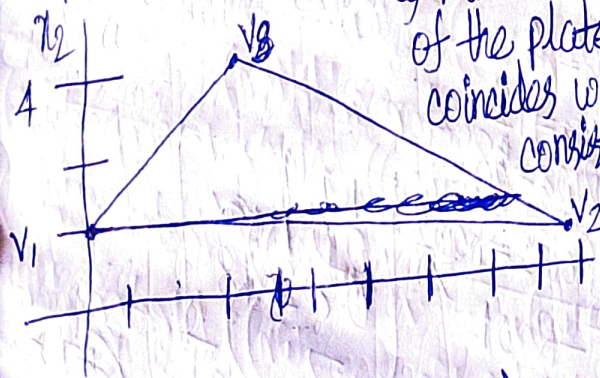
It's easy to see that $\forall i, i \in N, 1 \leq i \leq K, \left(\frac{m_i}{m}\right) \in \mathbb{R}$.

$\therefore \bar{V}$ is a linear combination of the vectors $V_1, \dots, V_K$.

By Defn, $\bar{V}$ is in $\text{Span}\{V_1, \dots, V_K\}$

31) A thin triangular plate of uniform density and thickness has vertices at $V_1 = (0,1)$, $V_2 = (8,1)$, and $V_3 = (2,4)$, as in the figure below, and the mass of the plate is 3gm.

a) Find the (x,y) coordinates of the center of mass of the plate. This "balance point" of the plate coincides with the center of mass of a system consisting of three 1-gram point masses located at the vertices of the plate.



Solution: $V_1 = (0,1)$ $V_2 = (8,1)$ $V_3 = (2,4)$ $m_1 = m_2 = m_3 = 1 \text{ gm}$
 $m = 3m_1 = 3 \text{ gm}$

$$\begin{aligned} \text{Center of mass} &= \frac{1}{3} [1(0,1) + 1(8,1) + 1(2,4)] = \frac{1}{3} \begin{bmatrix} 1 \cdot 0 + 1 \cdot 8 + 1 \cdot 2 \\ 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 4 \end{bmatrix} \quad \begin{matrix} \text{(Scalar multiplication)} \\ \text{(Vector addition)} \end{matrix} \\ &= \frac{1}{3} [(0,1) + (8,1) + (2,4)] = \frac{1}{3} [(0+8+2), (1+1+4)] \\ &= \frac{1}{3} [(10,6)] = \left(\frac{1}{3} \cdot 10, \frac{1}{3} \cdot 6\right) = \left(\frac{10}{3}, 2\right) \quad \text{(Scalar multiplication)} \end{aligned}$$

b) Determine how to distribute an additional mass of 6gm at the three vertices of the plate to move the balance point of the plate $(\frac{10}{3}, 2)$.

Solution: Let w_1, w_2 and w_3 denote the masses added at the three vertices, so that $w_1 + w_2 + w_3 = 6$.